

Enterprise Banking Microservices

A scalable and secure distributed banking backend

Course project — Database Systems Engineering & Distributed Backend Development

Team: Snigdha — 2520090160

Koumudi — 2520090161

Sumanjali — 2520090190

React.js → FastAPI → MySQL | JWT | Fraud Detection

Problem Statement

Traditional banking systems struggle with growth, security, and operational complexity.

- Scaling for high transaction volumes
- Secure authentication and authorization
- Consistent, reliable transaction processing
- Managing payments and account state across services
- Detecting suspicious/fraudulent behavior in real time

Project goal: build a secure, modular microservices platform managing customers, accounts, transactions, and fraud detection.

Project Objectives

Secure Architecture — Implement authentication, authorization, and secure REST APIs.

Scalability — Microservices design to scale individual capabilities independently.

Transaction Integrity — Consistent account balances and reliable transaction logging.

Fraud Detection — Identify suspicious patterns and flag high-risk activities.

User Experience — Clean frontend: registration, dashboard, transfers, and forms.

Literature Survey — Summary

Microservices Architecture — Improves modularity and scale; introduces distributed communication and consistency challenges.

Secure Authentication — Token-based auth and role-based access control are effective but need careful token lifecycle management.

Banking Transaction Systems — ACID semantics support correctness; scaling requires careful partitioning and compensating transactions.

Fraud Detection — Pattern-based and ML approaches detect anomalies; accuracy depends on data richness and labeled examples.

Research gap: Few works integrate scalable microservices, strong transactional guarantees, and inline fraud detection in a single platform.

Proposed System — Overview

We propose a microservices-based banking platform exposing secure REST APIs.

- Customer management and registration
- Authentication & authorization (token-based)
- Account and balance management
- Secure money transfers and payment recording
- Transaction ledger and audit trails
- Real-time fraud detection and alerting

Technology stack: React.js → FastAPI microservices → MySQL

System Architecture — Flow

Frontend (React) handles UI and client validation.

FastAPI services implement business logic per bounded context: auth, accounts, payments, and fraud.

Services communicate over REST (HTTP) and use MySQL for persistent state.

Fraud service subscribes to transaction events for near-real-time scoring.

Modules & Responsibilities

Customer Management — Profile CRUD, KYC fields, linkage to user accounts.

Authentication & Authorization — JWT tokens, refresh tokens, RBAC for operations.

Account Management — Account types, balance updates, overdraft rules.

Transfer & Payments — Atomic transfer flows with idempotency and audit logs.

Transaction Management — Durable transaction ledger and reconciliation endpoints.

Fraud Detection — Rule-based scoring and anomaly flags; alerts + quarantine actions.

Banking Dashboard — Aggregate views: balances, recent activity, alerts, KPIs.

Database Design — Conceptual

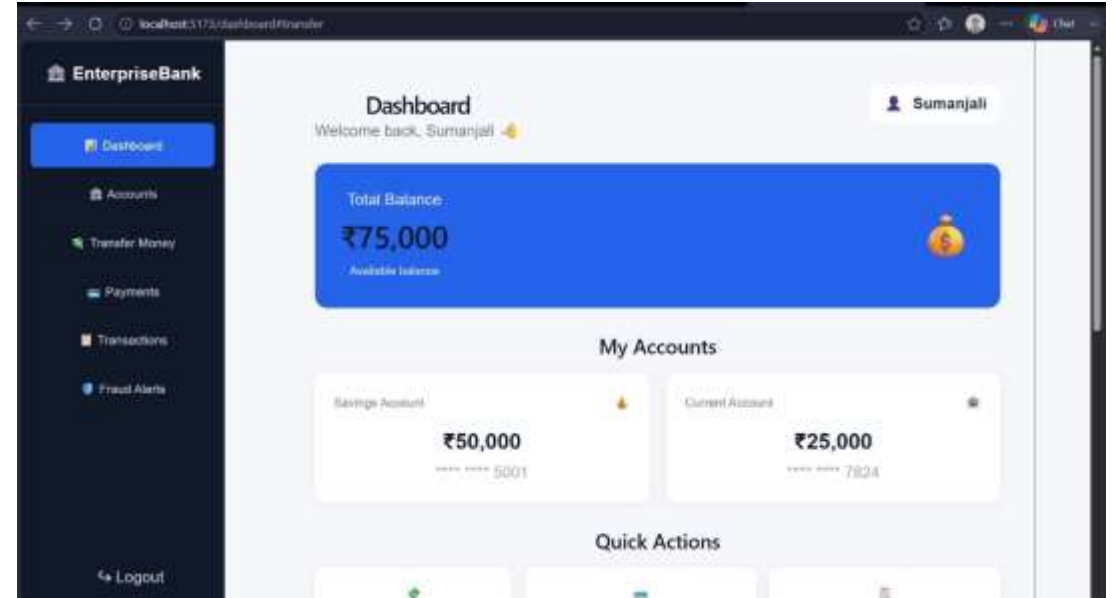
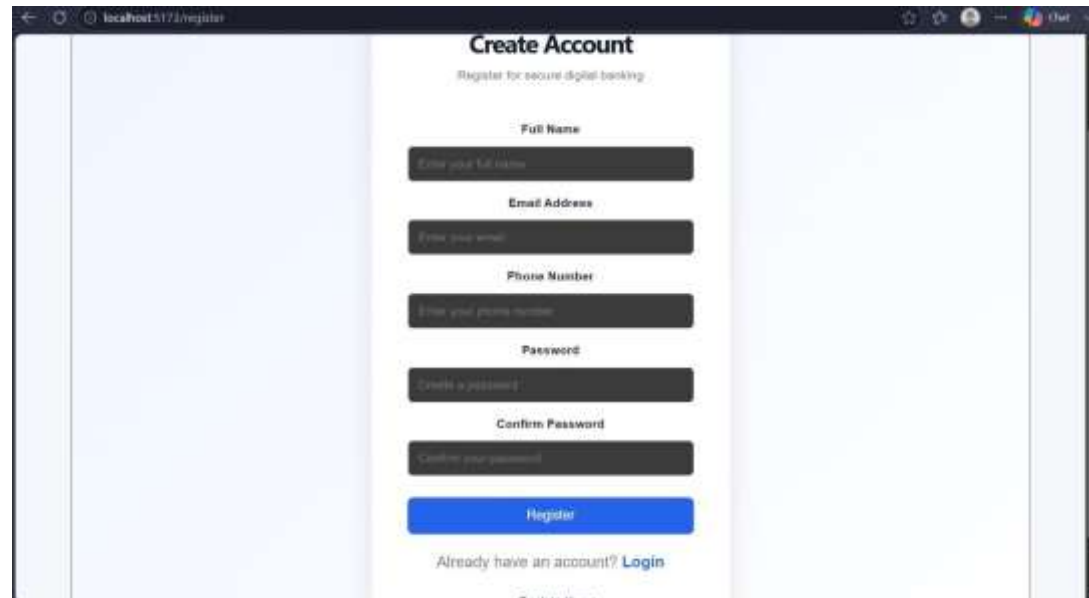
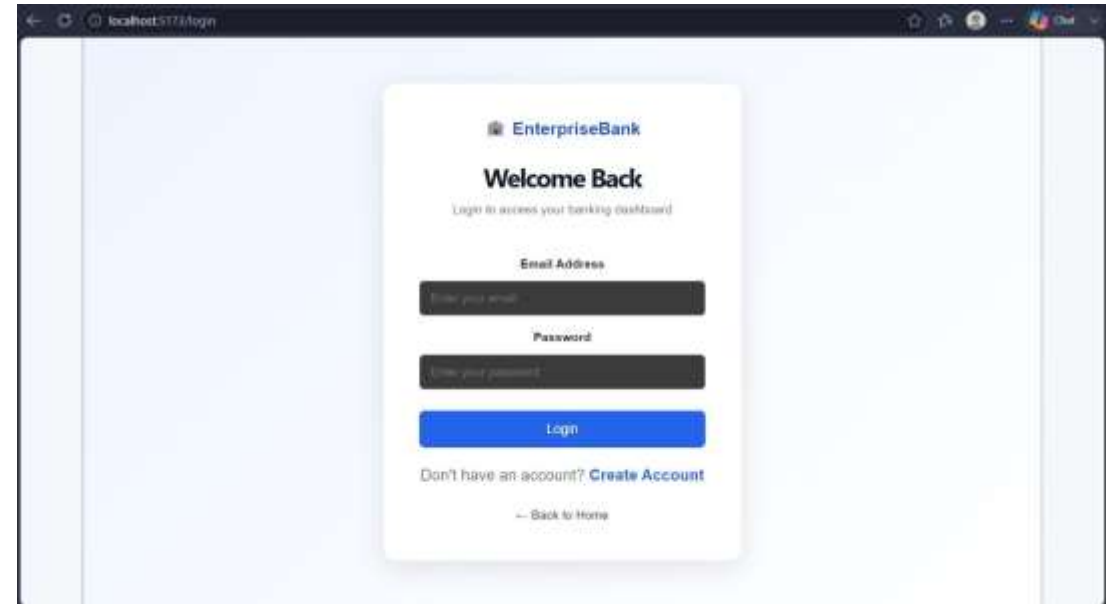
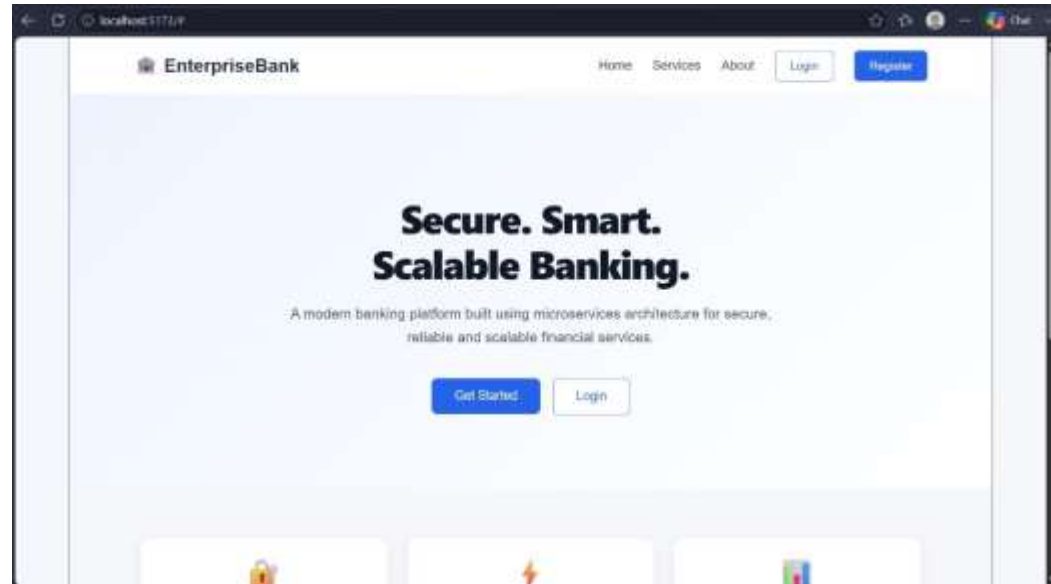
MySQL relational schema — main tables and key relationships:

- customers (customer_id, name, KYC, contact_info)
- users (user_id, customer_id, username, password_hash, roles)
- accounts (account_id, customer_id, balance, currency, status)
- transactions (tx_id, from_account, to_account, amount, status, timestamp)
- payments (payment_id, account_id, payee, amount, method, status)

Relationships: customers→users (1:N), customers→accounts (1:N), accounts→transactions (1:N).

Use indexes on account_id, customer_id, and transaction timestamps for performance.

WEBSITE



[← Back to Dashboard](#)



Transfer Money

Send money securely to another account

From Account

Recipient Account Number

Enter recipient account number

Transfer Amount

₹ Enter amount

Description

Enter transaction description

Continue Transfer



Fraud Risk Analysis

[← Back to Dashboard](#)



Fraud Detection

Monitor transactions and identify suspicious activity.

Transactions Analyzed

128

Suspicious Transactions

7

High Risk

2

Average Risk Score

18/100

Recent Risk Analysis

Transaction	Amount	Risk Score	Status
TXN001	₹5,000	12/100	LOW
TXN002	₹45,000	78/100	HIGH
TXN003	₹2,500	21/100	LOW
TXN004	₹30,000	65/100	REVIEW